

Comprehensive Whole-Genome and Structural Analysis of *Acinetobacter baumannii* SRR5891469: Antimicrobial Resistance, Virulence Factors, and OXA-23 Mutation Characterization

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Abstract

Background

Acinetobacter baumannii is an opportunistic Gram-negative pathogen associated with healthcare-associated infections and increasing levels of antimicrobial resistance. The emergence of carbapenem-resistant strains has become a major clinical concern worldwide. This study aimed to perform comprehensive genomic and structural characterization of *Acinetobacter baumannii* isolate SRR5891469, with particular emphasis on antimicrobial resistance determinants, virulence factors, and mutations within the OXA-23 beta-lactamase gene.

Methods

Whole-genome sequencing data (SRR5891469) were obtained from the NCBI Sequence Read Archive. Sequencing reads were subjected to quality assessment using FastQC, trimming with FastP, alignment against reference genome NZ_CP045110.1 using BWA-MEM, and variant calling using BCFtools. Genome annotation was performed using Prokka. Antimicrobial resistance genes and virulence factors were identified using ABRicate with CARD, NCBI, and VFDB databases. The OXA-23 gene was analyzed for mutations, and the

corresponding protein structure was predicted using SWISS-MODEL. Structural quality was evaluated using PROCHECK, ERRAT, and Verify3D.

Results

Genome annotation identified 3,790 genes, including 3,772 protein-coding genes and 3,699 coding sequences. Multiple antimicrobial resistance determinants were detected, including ADC-158, OXA-1117, ANT(3'')-IIa, sul2, and several multidrug efflux systems. A total of 110 virulence-associated genes were identified, primarily involved in biofilm formation, iron acquisition, pili assembly, and Type VI secretion systems. Mutation analysis of the OXA-23 beta-lactamase gene revealed multiple amino acid substitutions, including S64G, A96P, D134H, N146Y, Y153D, P170T, L171I, G240E, S246A, and K252N. Structural validation demonstrated high model quality, with 92.1% of residues located in favored Ramachandran regions, an ERRAT quality factor of 94.61, and 88.32% Verify3D compatibility.

Conclusion

The study provides a comprehensive genomic and structural assessment of *Acinetobacter baumannii* SRR5891469 and highlights the presence of multiple antimicrobial resistance determinants, virulence-associated genes, and OXA-23 mutations. The integration of genomic and structural analyses offers valuable insights into bacterial resistance mechanisms and pathogenicity.

Keywords:

Acinetobacter baumannii; Whole-genome sequencing; Antimicrobial resistance; OXA-23; Virulence factors; Structural bioinformatics; SWISS-MODEL

1. Introduction

Acinetobacter baumannii is a Gram-negative opportunistic pathogen that has emerged as one of the most important causes of healthcare-associated infections worldwide. The organism is frequently associated with bloodstream infections, ventilator-associated pneumonia, urinary tract infections, wound infections, and intensive care unit (ICU)-associated outbreaks. Its remarkable ability to survive under harsh environmental conditions and acquire multiple resistance determinants has made it a major public health concern.

The rapid emergence of multidrug-resistant (MDR) and extensively drug-resistant (XDR) *A. baumannii* strains has significantly limited available treatment options. Resistance to carbapenems, which are often considered last-resort antibiotics for severe bacterial infections, is particularly alarming. Carbapenem-resistant *A. baumannii* has been prioritized by the World Health Organization as a critical pathogen requiring urgent research and development of new therapeutic strategies.

Among the mechanisms responsible for carbapenem resistance, OXA-type beta-lactamases play a central role. The OXA-23 enzyme is one of the most widespread carbapenem-hydrolyzing class D beta-lactamases and has been frequently reported in clinical isolates worldwide. Mutations within resistance-associated genes may alter enzyme structure and function, potentially influencing substrate specificity, catalytic efficiency, and antimicrobial resistance phenotypes.

Advances in whole-genome sequencing and bioinformatics have enabled comprehensive characterization of bacterial pathogens at the genomic level. Genome analysis allows identification of antimicrobial resistance genes, virulence factors, and genetic variations associated with pathogenicity. Furthermore, protein structure prediction and validation provide valuable insights into the functional consequences of mutations and their potential impact on resistance mechanisms.

Therefore, the present study aimed to perform a comprehensive genomic and structural characterization of *Acinetobacter baumannii* isolate SRR5891469. The study focused on the identification of antimicrobial resistance determinants, virulence-associated genes, mutation analysis of the OXA-23 beta-lactamase gene, and structural assessment of the corresponding protein using computational modeling and validation approaches.

2. Materials and Methods

2.1 Dataset Retrieval

Whole-genome sequencing data of *Acinetobacter baumannii* SRR5891469 were obtained from the National Center for Biotechnology Information (NCBI) Sequence Read Archive (SRA). The sequencing dataset was generated using the Illumina HiSeq 2500 platform. The complete reference genome NZ_CP045110.1 was downloaded from the NCBI Genome database and used as the reference sequence for downstream analyses.

2.2 Quality Assessment and Read Processing

Raw sequencing reads were evaluated using FastQC to assess sequence quality, GC content, adapter contamination, and read distribution. Low-quality bases and adapter sequences were removed using FastP to obtain high-quality reads for downstream analyses.

2.3 Read Alignment and Variant Calling

Filtered reads were aligned against the reference genome NZ_CP045110.1 using BWA-MEM. The resulting alignment files were processed using SAMtools for sorting and indexing. Variant calling was performed using BCFtools to identify single nucleotide polymorphisms (SNPs) and insertion/deletion mutations (indels).

2.4 Genome Annotation

Genome annotation was performed using Prokka to identify coding sequences (CDSs), transfer RNA genes (tRNAs), ribosomal RNA genes (rRNAs), and other genomic features.

2.5 Antimicrobial Resistance Gene Identification

Antimicrobial resistance determinants were identified using ABRicate against the Comprehensive Antibiotic Resistance Database (CARD) and NCBI AMR databases.

2.6 Virulence Factor Identification

Virulence-associated genes were identified using ABRicate against the Virulence Factor Database (VFDB).

2.7 OXA-23 Mutation Analysis

The OXA-23 gene sequence was extracted from the annotated genome. Mutations were identified through comparative sequence analysis and validated using Integrative Genomics Viewer (IGV).

2.8 Protein Structure Prediction

The OXA-23 protein sequence was submitted to SWISS-MODEL for three-dimensional structure prediction.

2.9 Structural Validation

The quality of the predicted OXA-23 protein model was evaluated using PROCHECK, ERRAT, and Verify3D. Ramachandran plot analysis was used to assess stereochemical quality, while ERRAT and Verify3D were used to evaluate overall structural reliability and residue-environment compatibility.

Software and database versions used in this study included FastQC v0.12.1, FastP v0.24.0, BWA v0.7.17, SAMtools v1.21, BCFtools v1.21, Prokka v1.14.6, ABRicate v1.4.0, Integrative Genomics Viewer (IGV) v2.x, SWISS-MODEL Workspace, CARD, VFDB, and NCBI AMR databases.

3. Results

3.1 Genome Characterization

Whole-genome analysis of *Acinetobacter baumannii* SRR5891469 was performed using a reference-guided bioinformatics workflow. The sequencing reads were aligned against the reference genome NZ_CP045110.1, which has an approximate genome size of 3.98 Mb. Genome annotation using Prokka identified 3,790 genes, including 3,772 protein-coding genes and 3,699 coding DNA sequences (CDSs) distributed across a single circular chromosome.

The circular genome map generated from the annotated genome revealed the organization of coding sequences and genomic features throughout the chromosome (Figure 2). Annotation statistics are summarized in Table 2. The high number of coding sequences reflects the genomic complexity of *A. baumannii* and its ability to adapt to diverse environmental and clinical conditions.

The annotated genome served as the foundation for subsequent analyses, including antimicrobial resistance gene identification, virulence factor profiling, mutation analysis of the OXA-23 beta-lactamase gene, and structural characterization of resistance-associated proteins.

3.2 Antimicrobial Resistance Genes

Antimicrobial resistance gene profiling was performed using ABRicate against the CARD and NCBI AMR databases. A total of ten resistance-associated genes were identified within the genome of *Acinetobacter baumannii* SRR5891469 (Figure 3, Table 3).

The detected resistance determinants included beta-lactamase genes ADC-158 and OXA-1117, which are associated with resistance to beta-lactam antibiotics and carbapenems. The aminoglycoside resistance gene ANT(3'')-IIa and the sulfonamide resistance gene *sul2* were also identified. In addition, several multidrug efflux systems were detected, including AbaF, AbaQ, AmvA, AdeABC, AdeFGH, and AdeIJK.

The presence of multiple resistance determinants indicates a multidrug-resistant genetic profile and highlights the ability of *A. baumannii* to survive under antibiotic selection pressure. These findings are consistent with the global emergence of resistant *A. baumannii* strains in clinical settings.

3.3 Virulence Factors

Virulence-associated genes were identified using the Virulence Factor Database (VFDB). A total of 110 virulence-related genes were detected within the genome of *Acinetobacter baumannii* SRR5891469 (Figure 4, Table 4).

The identified virulence factors were primarily associated with biofilm formation, iron acquisition, pili assembly, Type VI secretion systems, and outer membrane proteins. Biofilm-associated genes contribute to surface attachment and persistence, while iron acquisition systems facilitate bacterial survival under iron-limited host environments. Pili-associated genes promote adhesion and colonization, whereas Type VI secretion system components are involved in bacterial competition and virulence.

The diverse virulence repertoire observed in this isolate suggests multiple mechanisms that contribute to pathogenicity, environmental persistence, and successful host colonization.

3.4 OXA-23 Mutation Analysis

The OXA-23 beta-lactamase gene was extracted from the annotated genome and subjected to mutation analysis. Several nucleotide substitutions were identified within the coding region and subsequently confirmed through visual inspection using Integrative Genomics Viewer (IGV) (Figure 5).

The corresponding amino acid substitutions included S64G, A96P, D134H, N146Y, Y153D, P170T, L171I, G240E, S246A, and K252N.

The analyzed gene corresponded to the annotated beta-lactamase OXA-23 (locus tag NJBJLHLK_01820) identified through Prokka genome annotation and located at genomic coordinates 1887843–1888667.

IGV analysis demonstrated strong read support for the detected variants, confirming the reliability of the variant-calling results. The identified mutations were further investigated through structural modeling and validation analyses.

3.5 Structural Modeling of OXA-23

The amino acid sequence of OXA-23 beta-lactamase was submitted to SWISS-MODEL for homology-based three-dimensional structure prediction. A high-confidence structural model was successfully generated (Figure 6).

The predicted structure displayed the characteristic fold of Class D beta-lactamases, including the conserved catalytic architecture required for hydrolysis of beta-lactam antibiotics. Structural analysis suggested that the identified amino acid substitutions were distributed throughout the protein and may contribute to functional variation in enzyme activity.

The generated model provided a structural framework for evaluating the potential impact of mutations on protein stability and antimicrobial resistance mechanisms.

3.6 Structural Validation

The quality of the predicted OXA-23 structural model was assessed using PROCHECK, ERRAT, and Verify3D (Figures 7–9).

Ramachandran plot analysis generated by PROCHECK demonstrated that 92.1% of residues were located within the most favored regions, 6.6% within additionally allowed regions, 0.8% within generously allowed regions, and only 0.4% within disallowed regions. These values indicate good stereochemical quality of the model.

ERRAT analysis produced an overall quality factor of 94.61, indicating a highly reliable structural model with acceptable non-bonded atomic interactions. Verify3D analysis further demonstrated that 88.32% of residues achieved acceptable three-dimensional to one-dimensional compatibility scores. The validation results are presented in Figures 7–9.

Collectively, these validation metrics confirm that the predicted OXA-23 structure is of high quality and suitable for structural interpretation of the identified mutations.

4. Discussion

4.1 Antimicrobial Resistance Profile

The antimicrobial resistance analysis revealed the presence of multiple resistance determinants within the genome of *Acinetobacter baumannii* SRR5891469. The identification of ADC-158 and OXA-1117 beta-lactamase genes indicates resistance mechanisms targeting beta-lactam antibiotics, including carbapenems. Additional resistance genes, such as ANT(3'')-IIa and sul2, further contribute to resistance against aminoglycosides and sulfonamides, respectively.

Acinetobacter baumannii has been designated as a critical priority pathogen by the World Health Organization due to the rapid emergence of multidrug-resistant and carbapenem-resistant strains.

The detection of several efflux pump systems, including AbaF, AbaQ, AmvA, AdeABC, AdeFGH, and AdeIJK, highlights the multidrug-resistant nature of the isolate. Efflux pumps play a critical role in reducing intracellular antibiotic concentrations and are commonly associated with treatment failure and persistence in clinical environments.

4.2 Virulence Determinants

Virulence factor profiling identified a diverse set of genes associated with pathogenicity and environmental adaptation. The presence of biofilm-associated genes suggests an enhanced ability to adhere to abiotic and biotic surfaces, contributing to persistence and chronic infection. Iron acquisition systems identified in the genome support bacterial survival under iron-limited host conditions.

Genes related to pili assembly, outer membrane proteins, and Type VI secretion systems further indicate the capacity of the organism to colonize host tissues, evade immune responses, and compete with other microorganisms. Collectively, these virulence determinants may contribute to the clinical success of *A. baumannii* as an opportunistic pathogen.

4.3 OXA-23 Mutation Analysis

Mutation analysis of the OXA-23 beta-lactamase gene revealed multiple amino acid substitutions distributed throughout the protein sequence. The observed substitutions, including S64G, A96P, D134H, N146Y, Y153D, P170T, L171I, G240E, S246A, and K252N, may influence protein conformation and enzymatic properties.

OXA-type beta-lactamases are among the most important mechanisms responsible for carbapenem resistance in *Acinetobacter baumannii*, and sequence variation within these enzymes may influence resistance phenotypes.

Although the functional consequences of these mutations require experimental validation, sequence variation within resistance-associated enzymes has previously been linked to altered substrate recognition, catalytic efficiency, and resistance phenotypes. Therefore, the identified mutations may contribute to the adaptive evolution of the enzyme.

4.4 Structural Modeling and Validation

Homology modeling using SWISS-MODEL generated a high-quality three-dimensional structure of the OXA-23 protein. The predicted model exhibited the characteristic fold of Class D beta-lactamases and provided a structural framework for assessing mutation distribution.

Ramachandran plot analysis showed that 92.1% of residues were located in the most favored regions, 6.6% in additionally allowed regions, 0.8% in generously allowed regions, and only 0.4% in disallowed regions. ERRAT analysis produced an overall quality factor of 94.61, while Verify3D indicated that 88.32% of residues achieved acceptable 3D–1D compatibility scores.

4.5 Biological Significance of the Findings

The combined genomic and structural analyses provide valuable insights into the resistance and virulence characteristics of *Acinetobacter baumannii* SRR5891469. The coexistence of multiple antimicrobial resistance determinants and virulence-associated genes suggests a highly adaptive organism capable of surviving under both antibiotic pressure and host-associated stress conditions.

Furthermore, structural characterization of the OXA-23 beta-lactamase offers a foundation for understanding the potential impact of sequence variation on protein function. These findings contribute to the growing body of knowledge regarding the molecular mechanisms underlying antimicrobial resistance and pathogenicity in *A. baumannii* and may support future efforts aimed at surveillance, therapeutic development, and infection control strategies.

One limitation of this study is that all analyses were performed using computational approaches and publicly available sequencing data. Experimental validation of the identified mutations and predicted structural effects would be required to confirm their biological significance. Future studies integrating phenotypic antimicrobial susceptibility testing and laboratory-based functional analyses would provide additional insights into the impact of these mutations on resistance mechanisms.

5. Conclusion

This study presents a comprehensive genomic and structural characterization of *Acinetobacter baumannii* SRR5891469 using whole-genome sequencing and bioinformatics approaches. Genome analysis identified multiple antimicrobial resistance determinants, including beta-lactamase genes, aminoglycoside resistance genes, sulfonamide resistance

genes, and multidrug efflux systems. Virulence profiling revealed numerous genes associated with biofilm formation, iron acquisition, pili assembly, and Type VI secretion systems, highlighting the pathogenic potential of the isolate.

Mutation analysis of the OXA-23 beta-lactamase gene identified several amino acid substitutions that may contribute to functional variation within the enzyme. Structural modeling of OXA-23 using SWISS-MODEL followed by validation through PROCHECK, ERRAT, and Verify3D demonstrated the reliability of the predicted protein structure.

Overall, the integration of whole-genome analysis, resistance profiling, virulence characterization, mutation analysis, and structural bioinformatics provides valuable insights into the mechanisms underlying antimicrobial resistance and pathogenicity in *Acinetobacter baumannii*. These findings contribute to the growing understanding of clinically important multidrug-resistant *Acinetobacter baumannii* strains and provide a foundation for future investigations into resistance evolution, surveillance, and therapeutic development.

Data Availability Statement

The complete project repository, analysis scripts, figures, tables, results, and supplementary materials are publicly available at:

GitHub Repository:

<https://github.com/Subhasree-2006/Acinetobacter-WGS-Analysis>

Archived Release (DOI):

<https://doi.org/10.5281/zenodo.20691969>

All data used in this study were obtained from publicly available resources, including the NCBI Sequence Read Archive (SRA) accession SRR5891469.

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